Guided ImMAP ascent:

MAP with side information, one conditional expectation at a time

ImMAP project notes

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Abstract

We want the MAP estimate of a target image $\boldsymbol{x}_0$ given all available side information: a guide image $\boldsymbol{x}_1$ (a paired scan, typically of another contrast), a measurement $\boldsymbol{y}$, or both. Revealing the side information one variable at a time splits the log-posterior exactly into the target prior plus one pointwise mutual information (PMI) term per variable, and each PMI term is a likelihood. The split survives Gaussian smoothing, and by Tweedie's formula the gradient of each term is the difference of two conditional expectations. So a single ascent step handles any set of side information. The chain matters in practice because it says how to *build* the needed expectation from the ones we can estimate: e.g. $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$ from a guide-conditioned network plus a data step. I^{2}SB, I^{2}SB-Recon, ImMAP2 and SDEdit are this one ascent with different conditioning sets, estimators and step schedules.

Setting. Real vectors in $\mathbb{R}^n$ (the complex case is identical with Wirtinger gradients). Pairs $(\boldsymbol{x}_0, \boldsymbol{x}_1)$ have a joint law; the measurement is $\boldsymbol{y} = \boldsymbol{E}\boldsymbol{x}_0 + \boldsymbol{n}$, $\boldsymbol{n} \sim \mathcal{N}(\boldsymbol{0}, \sigma_n^2 \boldsymbol{I})$, independent of $(\boldsymbol{x}_0, \boldsymbol{x}_1)$, so $\boldsymbol{y}$ and $\boldsymbol{x}_1$ are conditionally independent given $\boldsymbol{x}_0$. $\boldsymbol{u} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ independent of everything else; on the bridge it is the debiased state $(\boldsymbol{x}_t - t\boldsymbol{x}_1)/(1-t)$ at $\sigma^2 = Vt/(1-t)$. For a set $\boldsymbol{c} \subseteq \{\boldsymbol{x}_1, \boldsymbol{y}\}$ of conditioning variables,

$$D_{\boldsymbol{c}}(\boldsymbol{u}) \coloneqq \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{c}] \quad \text{at the current noise level } \sigma.$$

1 The objective

Given side information $\boldsymbol{c}$, the goal is

$$\hat{\boldsymbol{x}}_0 = \arg\max_{\boldsymbol{x}_0} \ \log p(\boldsymbol{x}_0 \mid \boldsymbol{c}), \qquad \boldsymbol{c} \in \{\varnothing,\ \{\boldsymbol{x}_1\},\ \{\boldsymbol{y}\},\ \{\boldsymbol{x}_1, \boldsymbol{y}\}\}. \tag{1}$$

We always condition on every independent source of information available. A variable that is a function of others adds nothing: e.g. the zero-filled image $\boldsymbol{E}^H\boldsymbol{y}$ is a function of $\boldsymbol{y}$, so conditioning on it as well as $\boldsymbol{y}$ is conditioning on $\boldsymbol{y}$. A guide scan is not, so $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1, \boldsymbol{y}) \neq p(\boldsymbol{x}_0 \mid \boldsymbol{y})$ in general.

2 Decomposing along a chain

Reveal the variables of $\boldsymbol{c}$ one at a time, in some order. With two variables, guide first:

$$\log p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1, \boldsymbol{y}) = \log p(\boldsymbol{x}_0) + \underbrace{\log \frac{p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)}{p(\boldsymbol{x}_0)}}_{\mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{x}_1)} + \underbrace{\log \frac{p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1, \boldsymbol{y})}{p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)}}_{\mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{y} \mid \boldsymbol{x}_1)} \tag{2}$$

$$= \log p(\boldsymbol{x}_0) + \log p(\boldsymbol{x}_1 \mid \boldsymbol{x}_0) + \log p(\boldsymbol{y} \mid \boldsymbol{x}_0) + \text{const}. \tag{3}$$

(2) is an identity: the ratios telescope. Each PMI term measures how much the newly revealed variable says about $\boldsymbol{x}_0$, given what was revealed before. In (3) each becomes a likelihood, by Bayes, $\mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{a} \mid \boldsymbol{b}) = \log p(\boldsymbol{a} \mid \boldsymbol{x}_0, \boldsymbol{b}) - \log p(\boldsymbol{a} \mid \boldsymbol{b})$, where the last term does not depend on $\boldsymbol{x}_0$, and by the conditional independence $p(\boldsymbol{y} \mid \boldsymbol{x}_0, \boldsymbol{x}_1) = p(\boldsymbol{y} \mid \boldsymbol{x}_0)$. So the objective is

$$\textit{target prior} + \textit{guide likelihood} + \textit{measurement likelihood},$$

whichever order the variables are revealed in, and dropping a variable drops its term. With one variable the chain has one step, and with none (1) is the prior alone.

The order does not change the objective. It changes the intermediate conditional expectations that appear below, i.e. which estimators the gradient is assembled from: guide-first uses $D_\varnothing, D_{\boldsymbol{x}_1}, D_{\boldsymbol{x}_1,\boldsymbol{y}}$; measurement-first uses $D_\varnothing, D_{\boldsymbol{y}}, D_{\boldsymbol{x}_1,\boldsymbol{y}}$.

3 Smoothing, and the gradient

As in ImMAP, we ascend a Gaussian smoothing of the objective while lowering the smoothing. Let $p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}) = \int \varphi_\sigma(\boldsymbol{u} - \boldsymbol{x})\, p(\boldsymbol{x} \mid \boldsymbol{c})\, d\boldsymbol{x}$, the law of the noisy target given $\boldsymbol{c}$, and

$$J_\sigma(\boldsymbol{u}) \coloneqq \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}), \qquad J_\sigma \to \log p(\cdot \mid \boldsymbol{c}) \text{ as } \sigma \to 0.$$

Proposition 1 (The chain survives smoothing; each term is a difference of expectations)**.** *Along the chain* $\varnothing \subset \boldsymbol{c}_1 \subset \cdots \subset \boldsymbol{c}_K = \boldsymbol{c}$*,*

$$J_\sigma(\boldsymbol{u}) = \log p_\sigma(\boldsymbol{u}) + \sum_{k=1}^{K} \log \frac{p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}_k)}{p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}_{k-1})}, \qquad \sigma^2 \nabla J_\sigma(\boldsymbol{u}) = \underbrace{D_\varnothing + \sum_{k=1}^{K} (D_{\boldsymbol{c}_k} - D_{\boldsymbol{c}_{k-1}})}_{= D_{\boldsymbol{c}}(\boldsymbol{u})} - \boldsymbol{u}, \quad (4)$$

and each ratio is the PMI between the noisy *target* $\boldsymbol{u}$ *and the* k*-th variable.*

Proof. The telescoping is the same as in (2), with $\boldsymbol{u}$ in place of $\boldsymbol{x}_0$; Bayes applies to $(\boldsymbol{u}, \boldsymbol{c}_k)$ because $\boldsymbol{\varepsilon}$ is independent of the side information. Tweedie's formula, $\sigma^2 \nabla \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}_k) = D_{\boldsymbol{c}_k}(\boldsymbol{u}) - \boldsymbol{u}$, holds for any such $\boldsymbol{c}_k$ (differentiate the convolution under the integral). Apply it to every term; the $-\boldsymbol{u}$'s cancel inside each ratio. □

What a variable contributes to the gradient is how far revealing it moves the conditional expectation, $D_{\boldsymbol{c}_k} - D_{\boldsymbol{c}_{k-1}}$; it vanishes when the variable is uninformative given the others. Summed, the chain gives back $D_{\boldsymbol{c}}$, so with an estimator of $D_{\boldsymbol{c}}$ itself nothing else is needed. The decomposition earns its keep when $D_{\boldsymbol{c}}$ is *not* directly available: then one estimates a less-conditioned expectation and adds the last increment separately. That is exactly what a data-consistency step does,

$$D_{\boldsymbol{x}_1,\boldsymbol{y}} = D_{\boldsymbol{x}_1} + \underbrace{(D_{\boldsymbol{x}_1,\boldsymbol{y}} - D_{\boldsymbol{x}_1})}_{\text{measurement term}},$$

with the increment estimated from the measurement model (Section 5).

The smoothed terms are not the unsmoothed likelihoods at a noisy image. The measurement term, for instance, involves $p(\boldsymbol{y} \mid \boldsymbol{u}, \boldsymbol{x}_1) = \int p(\boldsymbol{y} \mid \boldsymbol{x})\, p(\boldsymbol{x} \mid \boldsymbol{u}, \boldsymbol{x}_1)\, d\boldsymbol{x}$: the likelihood averaged over the targets the noisy state still allows. It is weak at large σ and sharpens to $p(\boldsymbol{y} \mid \boldsymbol{x}_0)$ as $\sigma \to 0$.

4 The step

Gradient ascent on J_σ with step size $h\sigma^2$ is, by (4),

$$\boxed{\boldsymbol{u}^+ = \boldsymbol{u} + h\sigma^2 \nabla J_\sigma(\boldsymbol{u}) = (1-h)\,\boldsymbol{u} + h\,\boldsymbol{v}, \qquad \boldsymbol{v} = D_{\boldsymbol{c}}(\boldsymbol{u}).} \tag{5}$$

By Tweedie's second-order identity, $-\nabla^2 J_\sigma = (\sigma^2 \boldsymbol{I} - \mathrm{Cov}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{c}])/\sigma^4 \preceq \sigma^{-2}\boldsymbol{I}$, so every $h \in (0, 2)$ increases J_σ (descent lemma), whatever $\boldsymbol{c}$ is.

The step runs inside the ImMAP continuation [1]: after each step, fresh noise is injected so that the iterate is again a noisy target at the next, smaller level. With the contraction $\sigma' = (1-\beta h)\sigma$, $\beta \in (0, 1]$, an accurate step leaves noise $(1-h)\sigma$, so the injection is

$$\gamma = \sigma\sqrt{(1-\beta h)^2 - (1-h)^2}. \tag{6}$$

$\beta = 1$ injects nothing (a deterministic continuation toward the MAP estimate); $\beta < 1$ samples the posterior.

Algorithm 1 Guided ImMAP ascent for $\max_{\boldsymbol{x}_0} \log p(\boldsymbol{x}_0 \mid \boldsymbol{c})$

Require: side information $\boldsymbol{c}$; an estimator of $D_{\boldsymbol{c}}(\boldsymbol{u}) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{c}]$ (directly, or assembled along a chain, (4)); start $\boldsymbol{u}_0$ at level σ_0; steps $h_k \in (0, 2)$; $\beta \in (0, 1]$; $\sigma_{\min}$

$\boldsymbol{u} \leftarrow \boldsymbol{u}_0$, $\sigma \leftarrow \sigma_0$
while $\sigma > \sigma_{\min}$ **do**
  $\boldsymbol{v} \leftarrow D_{\boldsymbol{c}}(\boldsymbol{u})$ ▷ $= \boldsymbol{u} + \sigma^2 \nabla J_\sigma(\boldsymbol{u})$
  $\boldsymbol{u} \leftarrow (1-h)\,\boldsymbol{u} + h\,\boldsymbol{v} + \gamma\,\boldsymbol{z}$, $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ ▷ step (5), injection (6)
  $\sigma \leftarrow (1-\beta h)\,\sigma$
end while
return $\boldsymbol{v}$

A natural start is the posterior mean, $D_{\boldsymbol{c}}$ at $\sigma = \infty$, i.e. $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{c}]$, re-noised to σ_0. An end-to-end reconstruction network (TGVN, LGGS) that regresses $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{y}, \boldsymbol{x}_1]$ in one shot is exactly that start: the first step of the ascent, without the rest.

5 Estimating the expectations

The algorithm consumes $D_{\boldsymbol{c}}$ and is indifferent to its source:

- $D_\varnothing$: a target-only denoiser, trained with the squared loss on $(\boldsymbol{u}, \boldsymbol{x}_0)$.
- $D_{\boldsymbol{x}_1}$: the paired bridge regressor, trained with the squared loss on bridge states and called on $\boldsymbol{x}_t = (1-t)\boldsymbol{u} + t\boldsymbol{x}_1$ at $t = \sigma^2/(\sigma^2 + V)$.
- $D_{\boldsymbol{x}_1,\boldsymbol{y}}$, learned: a bridge regressor that also sees $(\boldsymbol{y}, \boldsymbol{E})$ and the guide (SB-TGVN, BridgeLGGS).
- $D_{\boldsymbol{x}_1,\boldsymbol{y}}$ (or $D_{\boldsymbol{y}}$) from $D_{\boldsymbol{x}_1}$ (or $D_\varnothing$) by a **data step**: approximate the posterior given $\boldsymbol{u}$ and the other variables by $\mathcal{N}(D, \varsigma^2\boldsymbol{I})$ around the estimator's output D; with the Gaussian likelihood, the posterior mean given $\boldsymbol{y}$ too is the proximal point

$$(\boldsymbol{E}^H\boldsymbol{E} + \lambda\boldsymbol{I})^{-1}(\boldsymbol{E}^H\boldsymbol{y} + \lambda D), \qquad \lambda = \sigma_n^2/\varsigma^2, \tag{7}$$

  solved by CG warm-started at D. Since the posterior variance is at most σ^2 on average ($\mathbb{E}\|\boldsymbol{x}_0 - D\|^2 = \mathrm{mmse}(\sigma) \le n\sigma^2$), the principled weight follows the noise level, $\lambda(\sigma) = \sigma_n^2/(c\sigma^2)$, $c \le 1$.

6 Worked case: standard I^{2}SB

The source image $\boldsymbol{x}_1$ is the only side information; there is no measurement.

The objective. $\boldsymbol{c} = \{\boldsymbol{x}_1\}$, and the chain $\varnothing \subset \{\boldsymbol{x}_1\}$ has one step:

$$\max_{\boldsymbol{x}_0} \log p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1) = \log p(\boldsymbol{x}_0) + \underbrace{\log p(\boldsymbol{x}_1 \mid \boldsymbol{x}_0)}_{\mathrm{PMI}(\boldsymbol{x}_0;\boldsymbol{x}_1)+\mathrm{const}}.$$

A target that is plausible on its own and consistent with this particular source. Smoothed and differentiated (Proposition 1),

$$\sigma^2 \nabla J_\sigma(\boldsymbol{u}) = \underbrace{(D_\varnothing - \boldsymbol{u})}_{\text{prior pull}} + \underbrace{(D_{\boldsymbol{x}_1} - D_\varnothing)}_{\text{what } \boldsymbol{x}_1 \text{ adds}} = D_{\boldsymbol{x}_1}(\boldsymbol{u}) - \boldsymbol{u}.$$

I^{2}SB never splits the two pulls: its network estimates their sum, $D_{\boldsymbol{x}_1}$, directly.

What the network estimates. Training draws a pair $(\boldsymbol{x}_0, \boldsymbol{x}_1)$, a time $t \sim \mathcal{U}[0,1]$ and a bridge state

$$\boldsymbol{x}_t = (1-t)\,\boldsymbol{x}_0 + t\,\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(t)\,\boldsymbol{\varepsilon}, \qquad \sigma_{\mathrm{sb}}(t)^2 = Vt(1-t), \quad V = 4\tau^2,$$

and regresses $\boldsymbol{x}_0$ with the squared loss, so $D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t, \boldsymbol{x}_1]$. Removing the known $\boldsymbol{x}_1$ part of the state turns it into a noisy target,

$$\boldsymbol{u} = \frac{\boldsymbol{x}_t - t\,\boldsymbol{x}_1}{1-t} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}, \qquad \sigma^2 = \frac{Vt}{1-t},$$

and $(\boldsymbol{x}_t, \boldsymbol{x}_1) \mapsto (\boldsymbol{u}, \boldsymbol{x}_1)$ is invertible, so $D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1] = D_{\boldsymbol{x}_1}(\boldsymbol{u})$ at $\sigma(t)$. *Calling the network at bridge time t evaluates the gradient of the objective smoothed at $\sigma(t)$:* $\sigma(t)^2 \nabla J_{\sigma(t)} = D_\theta - \boldsymbol{u}$. The bridge time is the smoothing level, from $\sigma = 0$ at $t = 0$ to $\sigma = \infty$ at $t = 1$.

One step, in both coordinates. From time t to $s < t$, with $r = \sigma(s)/\sigma(t)$, i.e. $r^2 = \frac{s(1-t)}{t(1-s)}$, I^{2}SB takes the step (5) with $h = 1 - r^2$ and the injection (6) with $\beta = 1/(1+r)$:

$$\boldsymbol{u}_s = \boldsymbol{u}_t + (\sigma_t^2 - \sigma_s^2)\,\nabla J_{\sigma_t}(\boldsymbol{u}_t) + \sigma_s\sqrt{1-r^2}\,\boldsymbol{z} = r^2\,\boldsymbol{u}_t + (1-r^2)\,D_{\boldsymbol{x}_1}(\boldsymbol{u}_t) + \sigma_s\sqrt{1-r^2}\,\boldsymbol{z}.$$

The step size is the variance removed, and the injection tops the leftover noise $r^2\sigma_t = r\sigma_s$ back up to exactly σ_s. Mapped back with $\boldsymbol{x}_s = (1-s)\boldsymbol{u}_s + s\,\boldsymbol{x}_1$, the $\boldsymbol{x}_1$ terms cancel and

$$\boldsymbol{x}_s = \tfrac{s}{t}\,\boldsymbol{x}_t + \left(1 - \tfrac{s}{t}\right)\hat{\boldsymbol{x}}_0 + \sqrt{V\tfrac{s(t-s)}{t}}\,\boldsymbol{z}, \qquad \hat{\boldsymbol{x}}_0 = D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1),$$

which is I^{2}SB's reverse step: the bridge posterior between $\boldsymbol{x}_t$ and the predicted $\hat{\boldsymbol{x}}_0$.

A run, from $\boldsymbol{x}_1$ to $\boldsymbol{x}_0$. On a ladder $1 = t_0 > t_1 > \cdots > t_K = 0$:

- **First step** ($t_0 = 1$, $\boldsymbol{x}_t = \boldsymbol{x}_1$, $\sigma = \infty$, $r = 0$, $h = 1$). The state carries no information beyond $\boldsymbol{x}_1$, so the network returns the posterior mean $\hat{\boldsymbol{x}}_0 = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1]$: a blurry average of every target consistent with $\boldsymbol{x}_1$. The step jumps to it and re-noises: $\boldsymbol{x}_{t_1} = (1-t_1)\,\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1] + t_1\boldsymbol{x}_1 + \sigma_{\mathrm{sb}}(t_1)\boldsymbol{z}$. This is a warm start at noise level $\sigma(t_1)$, not "starting at $\boldsymbol{x}_1$".

- **Middle steps** ($0 < h < 1$). Each moves the state the fraction $1 - r^2$ toward the current estimate and re-injects noise. The fresh noise commits the run to one of the targets the estimate was averaging over; the next estimate, at lower σ, is sharper. Unrolled, the state is a precision-weighted average of all estimates so far.
- **Last step** ($s = 0$, $\sigma = 0$, $r = 0$, $h = 1$). No noise is injected and the output is the final estimate $\hat{\boldsymbol{x}}_0 = D_\theta(\boldsymbol{x}_{t_{K-1}}, t_{K-1}, \boldsymbol{x}_1)$.

step $t \to s$	$\sigma(t)$	$\sigma(s)$	r	$h = 1 - r^2$	β	injected (in $\boldsymbol{u}$)	$\boldsymbol{x}_s$: weight on $\boldsymbol{x}_t$, $\hat{\boldsymbol{x}}_0$	noise (in $\boldsymbol{x}$)
$1 \to 0.75$	∞	0.658	0	1	1	0.658	0.75, 0.25	0.165
$0.75 \to 0.5$	0.658	0.380	0.577	0.667	0.634	0.310	0.67, 0.33	0.155
$0.5 \to 0.25$	0.380	0.219	0.577	0.667	0.634	0.179	0.50, 0.50	0.134
$0.25 \to 0$	0.219	0	0	1	1	0	0, 1	0

Table 1: Four uniform steps at $\tau = 0.19$ ($V = 0.144$). At the first step the weight 0.75 falls on $\boldsymbol{x}_t = \boldsymbol{x}_1$, i.e. $\boldsymbol{x}_{t_1} = 0.25\,\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1] + 0.75\,\boldsymbol{x}_1 + 0.165\,\boldsymbol{z}$. Uniform steps in t give equal ratios r in the middle of the ladder.

A sample, not the MAP estimate. $\beta = 1/(1 + r) < 1$ re-injects noise, so I^{2}SB returns (approximately) a sample of $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)$. The approximation is the point estimate: the exact reverse step would also carry the spread $(1 - r^2)^2\,\mathrm{Cov}[\boldsymbol{x}_0 \mid \boldsymbol{u}_t, \boldsymbol{x}_1]$ of the endpoint, which vanishes on fine ladders. The same network gives the deterministic continuation toward the MAP estimate with $\beta = 1$, i.e. $h = 1 - r$ and no injection: the Euler step of the probability-flow ODE. Keeping $h = 1 - r^2$ and only dropping the noise (`deterministic=True`) is neither: the state then lands at noise $r^2\sigma_t < \sigma_s$, below the level the next call is told.

7 The methods as instances

Every method below is Algorithm 1. What differs is the conditioning set (what is estimated), the estimator, and the hyperparameters: noise ladder, h_k, β, start.

I^{2}SB [2]. $\boldsymbol{c} = \{\boldsymbol{x}_1\}$, $\boldsymbol{v} = D_{\boldsymbol{x}_1}$, the paired bridge regressor; fixed ladder $\sigma_k = \sigma(t_k)$ from ∞ to 0 on NFE uniform bridge steps; with $r_k = \sigma_{k+1}/\sigma_k$,

$$h_k = 1 - r_k^2, \qquad \beta_k = \frac{1}{1 + r_k}, \qquad \gamma_k^2 = \sigma_{k+1}^2(1 - r_k^2) \tag{8}$$

(Section 6).

I^{2}SB-Recon and I^{2}SB-Inversion [3]. The guide is a fully sampled scan of *another contrast* (T1 for a T2 target; PD for PDFS here), and it is the bridge start: $\boldsymbol{x}_1 =$ guide. The objective is the full posterior, $\boldsymbol{c} = \{\boldsymbol{x}_1, \boldsymbol{y}\}$:

$$\log p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1, \boldsymbol{y}) = \log p(\boldsymbol{x}_0) + \log p(\boldsymbol{x}_1 \mid \boldsymbol{x}_0) + \log p(\boldsymbol{y} \mid \boldsymbol{x}_0) + \text{const.}$$

The network only estimates $D_{\boldsymbol{x}_1}$ (it never sees $\boldsymbol{y}$); the measurement term is added by the data step (7) from $D_{\boldsymbol{x}_1}$, with $\lambda = 7 \cdot 10^{-5}$ fixed and 7 CG iterations, giving $\boldsymbol{v} \approx D_{\boldsymbol{x}_1, \boldsymbol{y}}$. Since the guide is a separate scan, this term is genuinely informative: the data step is the only place the measurement enters. Hyperparameters as I^{2}SB (8), with 80 steps. I^{2}SB-Inversion then walks the reconstruction back up the bridge deterministically and runs the same ascent a second time from there.

ImMAP2 (`diffusion/immap.py`). No guide: $\boldsymbol{c} = \{\boldsymbol{y}\}$, $\boldsymbol{v} \approx D_{\boldsymbol{y}}$, the data step (7) applied to a target-only denoiser $D_\varnothing$, with $\lambda_k = \lambda_0 \sigma_n^2 (1+\sigma_k^2)/\sigma_k^2$ (posterior variance $\sigma_k^2/(1+\sigma_k^2)$, the ΠGDM rule) solved to tolerance. The ladder is self-paced, $\sigma_k^2 = \frac{1}{n}\|D_\varnothing(\boldsymbol{u}_k) - \boldsymbol{u}_k\|^2$ (slightly low on average: $\mathbb{E}\|D_\varnothing - \boldsymbol{u}\|^2 = n\sigma^2 - \text{mmse}$); $h_k = h_0 k/(1+h_0(k-1))$ with $h_0 = 0.01$, $\beta = 0.05$; start mean$(\boldsymbol{E}^H\boldsymbol{y}) + \boldsymbol{z}$ at $\sigma = 1$. The implementation returns the final state $\boldsymbol{u}$, not $\boldsymbol{v}$.

SDEdit. $\boldsymbol{c} = \varnothing$, $\boldsymbol{v} = D_\varnothing$: the source image enters only as the start $\boldsymbol{u}_0 = \boldsymbol{x}_1 + \sigma_0 \boldsymbol{z}$, not as conditioning.

A zero-filled bridge start. If $\boldsymbol{x}_1 = \boldsymbol{E}^H\boldsymbol{y}$ (our `sbunet` baseline), $\boldsymbol{x}_1$ is a function of $\boldsymbol{y}$, so $\boldsymbol{c} = \{\boldsymbol{x}_1, \boldsymbol{y}\}$ is the same information as $\{\boldsymbol{y}\}$: the target is the unguided posterior $p(\boldsymbol{x}_0 \mid \boldsymbol{y})$. The data step then adds no information in exact arithmetic and only corrects the network's errors.

method	conditions on	estimator of $D_{\boldsymbol{c}}$	ladder, h, β
SDEdit	nothing	$D_\varnothing$	free
ImMAP2	$\boldsymbol{y}$	data step on $D_\varnothing$, $\lambda_k \propto (1+\sigma_k^2)/\sigma_k^2$	self-paced; KS h_k; $\beta = 0.05$
I^{2}SB	guide	$D_{\boldsymbol{x}_1}$ (bridge regressor)	fixed ladder; $h = 1 - r^2$; $\beta = \frac{1}{1+r}$
I^{2}SB-Recon / -Inversion	guide, $\boldsymbol{y}$	data step on $D_{\boldsymbol{x}_1}$, $\lambda = 7 \cdot 10^{-5}$	as I^{2}SB, 80 steps
SB-TGVN, BridgeLGGS (any start)	guide, $\boldsymbol{y}$	learned $D_{\boldsymbol{x}_1,\boldsymbol{y}}$ (the net sees both)	as I^{2}SB
`sbunet` (zero-filled start)	$\boldsymbol{y}$	$D_{\boldsymbol{x}_1}$ with $\boldsymbol{x}_1 = \boldsymbol{E}^H\boldsymbol{y}$, + data step	as I^{2}SB
TGVN, LGGS end-to-end	guide, $\boldsymbol{y}$	learned $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{y}, \boldsymbol{x}_1]$	one step from $\sigma = \infty$

Table 2: Every method is Algorithm 1. The first two columns say what is estimated; the last is how the ascent is run.

Two consequences. First, I^{2}SB-Recon and ImMAP2 differ in what they condition on (the guide) as well as in the estimator and hyperparameters, so comparing them conflates the three; the table says which single change isolates each. Giving ImMAP2 the guide means replacing $D_\varnothing$ by $D_{\boldsymbol{x}_1}$; giving I^{2}SB-Recon ImMAP2's λ_k isolates the data weight. Second, for SB-TGVN the bridge start is a parameterisation, not an objective: whatever $\boldsymbol{x}_1$ is, the net conditions on the guide and $\boldsymbol{y}$, so the three starts target the same posterior and differ only in the path.

References

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